SESSION 22

SQL with Python & Jupyter Notebook

Q-1: Use pandas read_sql() to load all records from *restaurants* table into a DataFrame

```
import sqlite3
import pandas as pd

# Connect to local SQLite database
conn = sqlite3.connect("my_database.db")

# Load all records into a DataFrame
df = pd.read_sql("SELECT * FROM restaurants;", conn)

# Display first 5 rows
print(df.head())
```

Q-2: Run a SQL query in Jupyter Notebook using %sql magic

```
%load_ext sql
%sql sqlite:///my_database.db

# Select movies with rating above 8
%sql SELECT * FROM movies WHERE rating > 8;
```

Q-3: Combine SQL and Python to analyze Zomato-style order data

```
# Load last 100 orders
orders_df = pd.read_sql("SELECT * FROM orders ORDER BY order_date DESC LIMIT 100;", conn)

# Find top 3 most ordered food items
top_items = orders_df['food_item'].value_counts().head(3)
print(top_items)
```

Q-4: Create a bar chart showing matches played by each team

```
import matplotlib.pyplot as plt
import seaborn as sns

# Load IPL matches data
matches_df = pd.read_sql("SELECT * FROM matches;", conn)

# Count matches per team
team_counts = matches_df['team'].value_counts()

# Plot bar chart
sns.barplot(x=team_counts.index, y=team_counts.values)
plt.xticks(rotation=45)
plt.title("Number of Matches Played by Each Team")
plt.show()
```

Q-5: Jupyter Notebook cell to fetch users with more than 5 orders

```
import sqlite3
import pandas as pd

# Connect to SQLite database
conn = sqlite3.connect("my_database.db")

# SQL query to fetch users with more than 5 orders
query = """
SELECT user_id, COUNT(order_id) AS total_orders
FROM orders
GROUP BY user_id
HAVING COUNT(order_id) > 5;
"""

# Load result into DataFrame
users_df = pd.read_sql(query, conn)
```